

Literature Survey

VULNEX – AI-Powered Website Vulnerability Assessment Platform

1. Introduction

Website security assessment has become a critical concern as organizations increasingly depend on web infrastructure, with a large share of security incidents traced to misconfigured HTTP headers, weak SSL/TLS deployments, and unpatched vulnerabilities. Existing research addresses these concerns in isolated fragments – either evaluating scanning tools, assessing OWASP-based vulnerabilities, or applying machine learning to predict risk – without combining scanning, AI-based scoring, OWASP mapping, and automated reporting into one platform. This survey reviews three representative studies drawn from IEEE Xplore and ScienceDirect to identify the gap that VULNEX addresses.

2. Review of Existing Work

2.1 Vulnerability Assessment and Penetration Testing of Web Applications

Source: IEEE Xplore. Nagpure and Kurkure (2017) examine web applications vulnerable to SQL Injection, Cross-Site Scripting, CSRF, and Security Misconfiguration, framed against the OWASP Top 10, and compare vulnerability assessment with penetration testing as complementary techniques. The work is methodology-focused and does not propose an automated scoring engine or an AI-based risk predictor.

2.2 Evaluation of Web Vulnerability Scanners Based on the OWASP Benchmark

Source: IEEE Xplore. Mburano and Si (2018) benchmark open-source web vulnerability scanners against the OWASP Benchmark test suite, comparing detection accuracy and false-positive rates across tools. The study is limited to comparing existing scanners and does not integrate results into a unified security score, a predicted risk level, or a downloadable report.

2.3 Machine-Learning-Based Cyber Risk Prediction

Source: ScienceDirect (Expert Systems with Applications). Kia, Murphy, Sheehan, and Shannon (2024) propose a machine learning model that uses CVE (Common Vulnerabilities and Exposures) data to predict and rank cyber risk scores. This demonstrates that vulnerability data can be converted into a quantitative risk output using ML, but the scope is limited to CVE-level risk ranking rather than live scanning of a website's headers, SSL/TLS, or cookie configuration.

3. Comparative Summary

Author(s), Year & Source	Focus	Limitation w.r.t. VULNEX
Nagpure & Kurkure (2017) <i>IEEE Xplore</i>	VA vs. penetration testing methodology, OWASP-aligned	No automated scoring, AI risk prediction, or reporting
Mburano & Si (2018) <i>IEEE Xplore</i>	Scanner benchmarking against OWASP Benchmark	Accuracy comparison only; no header/SSL/cookie analysis
Kia et al. (2024) <i>ScienceDirect</i>	ML-based cyber risk prediction from CVE data	CVE-level ranking, not live website configuration scanning

4. Research Gap and Positioning of VULNEX

The reviewed studies show that scanner evaluation, OWASP-based assessment, and ML-based risk prediction have each been studied separately, but no reviewed work combines live website scanning (headers, SSL/TLS, cookies) with AI-based risk scoring, OWASP-category mapping, scan history, and automated PDF reporting in one accessible platform. VULNEX addresses this gap by scanning a user-registered website, applying a

machine learning model – in the spirit of Kia et al. (2024) – to generate a security score and risk level, mapping findings to OWASP categories as in Nagpure and Kurkure (2017), and benchmarking detection logic against standards similar to Mburano and Si (2018), while adding scan history tracking and downloadable reports that none of the reviewed systems provide together.

References

- [1] P. Nagpure and S. Kurkure, “Vulnerability Assessment and Penetration Testing of Web Application,” in *Proc. 2017 Int. Conf. on Computing, Communication, Control and Automation (ICCUBEA)*, IEEE, 2017. Available: <https://ieeexplore.ieee.org/document/8463920>
- [2] B. Mburano and W. Si, “Evaluation of Web Vulnerability Scanners Based on OWASP Benchmark,” in *Proc. 2018 26th Int. Conf. on Systems Engineering (ICSEng)*, IEEE, 2018, pp. 1–6. Available: <https://ieeexplore.ieee.org/document/8638176>
- [3] A. N. Kia, F. Murphy, B. Sheehan, and D. Shannon, “A Cyber Risk Prediction Model Using Common Vulnerabilities and Exposures,” *Expert Systems with Applications*, vol. 237, 121599, 2024. Available: <https://www.sciencedirect.com/science/article/pii/S0957417423021012>